

at a fast rate and we need to match this pace of demand and supply. Open source initiatives will be playing a major role in building an open ecosystem to enable innovations, implementing them in products and to meet the global demand for consumer products. These are some of the areas that I've been trying to cater to through the Beagle Board initiative."

Open source software is in fact embedded into the board—it consists of a boot loader, various libraries, Linux kernel and applications running on top of it.

What's in a board?

The revision C of Beagle Board was introduced in May 2009—in India, Cranes Software International is the official distributor. So, coming back to our real question: what can a developer look for in the board?

First, the OMAP3530 processor. OMAP3530 is a high end, multi-core, application processor with ARM Cortex A8, Generic C64x DSP and an SGX 3D graphics accelerator. With the ARM running at 600 MHz, DSP at 430 MHz, SGX at 111MHz, the effective total system power becomes 1.2GHz. This gives developers an opportunity to explore a variety of desktop applications on hand-held devices with much lower power consumption compared to any other x86 or Intel Atom-based devices. What's even more interesting is that the board can run desktop

applications—the OpenOffice.org office suite, GIMP, Firefox, etc—with only 2 watts of power consumption. With respect to heat dissipation, the OMAP3530 processor is rather superior. There's no need for any sort of cooling device (like a heat sink or a fan) as the temperature remains below 50 degrees Celsius. Now, that's what we call an ideal green computing device.

Although there are a lot of interfaces available on the board, the high speed USB and USB OTG are worth mentioning first. A single high-speed USB on the board can be used to connect 128 devices with the help of a hub. However, when we use a high speed USB port, the USB host acts only as the master and the device acts as a slave. A USB OTG allows the host to behave either as a master or a slave, but not at the same time. When the USB OTG is behaving as a host, it should be connected to a 5V power supply. But, when connected to a PC, it behaves as a slave.

It's no surprise that the device, according to the manufacturers, is aimed at exploring and developing applications on OpenGL, OpenMax, OpenOffice.org, Open CV, Browsers, 3D gaming, GStreamer, 3D UI, etc. But would the device be able to run all these applications in parallel? In which case, how much power will the device typically consume? Would a cell phone-like battery be enough to support it for hours? Also, how much RAM can be fitted into the board to take care of intensive multi-tasking?

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